Project Goal

Given a group G and a generating set S, we construct the Cayley graph

$$\Gamma = \mathrm{Cay}(G, S),$$

and study whether this graph can be regarded as the **1-skeleton of a closed surface** with oriented condition which means each neiborhood cycle has converse oiented.

The key idea is choosing a **G-invariant face structure** coming from relator-induced cycles. After that, we compute

$$\chi = V - E + F$$

to determine the topological type of the resulting surface. This is exactly the framework used in the D_4 notes and code: first construct the Cayley graph and candidate faces combinatorially, then check edge incidences, compute Euler characteristic, and finally build a geometric visualization if desired.

General Workflow

1. **Choose a group and corresponding n-cycle(G, S)**
2. **Construct the Cayley graph $\Gamma = \mathrm{Cay}(G, S)$**
3. **Choose a based cycle from a relator word**
4. **Take all G-translates to form a G-invariant cycle set**
5. **Check neiborhood cycle has converse oiented**
6. **Check whether every edge belongs to exactly two faces and with inverse direction**
7. **Compute $\chi = V - E + F$**
8. **Determine the surface type**
9. **Construct a geometric embedding / visualization**

Meaning of Each Step

Step 1. Choose (G, S)

We first choose a group G and a generating set S. This determines the basic combinatorial structure of the Cayley graph.

- Vertices are group elements.
- Edges come from left multiplication by generators.
- If $s \in S$, the corresponding directed edge is $g \to sg$.

Step 2. Construct the Cayley graph

We build $\Gamma = \mathrm{Cay}(G, S)$. At this stage, we already know:

- The number of vertices V.
- The adjacency structure.
- The degree of each vertex.

Step 3. Choose a based cycle from a relator word

A face should come from a word in the generators that closes up. If $s_1 s_2 \dots s_k = e$, then starting from the identity gives a closed walk:

$$C = (e,\ s_1,\ s_2 s_1,\ \dots,\ e)$$

This is called a **based cycle**.

Step 4. Take all G-translates

Since we want the face structure to respect the symmetry of the group, we must include all of its left translates:

$$\mathcal{O}(C) = \{\, gC \mid g \in G \,\}$$

This produces a G-**invariant cycle set**.

Step 5. Check edge usage

We count how many times each edge appears among all candidate faces.

1. **Success:** Every edge appears exactly twice $\rightarrow$ Closed surface.
2. **Failure:** Some edge appears > 2 times $\rightarrow$ Too many faces meet.
3. **Incomplete:** Some edge appears ≤ 1 time $\rightarrow$ More face orbits needed.

Step 6. Compute Euler characteristic

If the edge condition is satisfied:

$$\chi = V - E + F$$

Step 7. Determine the surface type

For a closed orientable surface, $\chi = 2 - 2g$.

- $\chi = 2$: Sphere type
- $\chi = 0$: Torus type
- $\chi < 0$: Higher genus surface

Step 8. Construct a geometric embedding

- **Sphere ($\chi = 2$):** Polyhedral realization.
- **Torus ($\chi = 0$):** Torus parameterization or spectral embedding.

Role of the Two Existing Codes

Code A: Combinatorial / Topological Checker

- Defines group and generators.
- Traces relator words and deduplicates cycles.
- **Answers:** Does this system define a closed surface? What is the χ?

Code B: Geometric Torus Visualization

- Used after torus-compatibility is confirmed.
- Uses torus parameterization:

$$f(u,v) = ((R + r\cos v)\cos u,\ (R + r\cos v)\sin u,\ r\sin v)$$

- **Answers:** How do we draw it in 3D?

Examples

Example 1: D_4 (Cube-like)

- **Generators:** $S = \{r, r^{-1}, s\}$
- **Cycles:** $r^4 = e$, $(rs)^2 = e$
- **Stats:** $V = 8, E = 12, F = 6$
- **Result:** $\chi = 2$ (**Sphere**)

Example 2: D_4 (Hexagon)

- **Based Cycle:** $C = (1, r, r^2, r^2s, rs, s, 1)$
- **Stats:** $V = 8, E = 12, F = 4$
- **Result:** $\chi = 0$ (**Torus**)

Example 3: S_3

- **Generators:** $s = (12), t = (23)$
- **Stats:** $V = 6, E = 6, F = 2$
- **Result:** $\chi = 2$ (**Sphere**)

Example 4: S_4

- **Generators:** s_1, s_2, s_3 (Coxeter)
- **Stats:** $V = 24, E = 36, F = 14$
- **Result:** $\chi = 2$ (**Sphere/Permutahedron**)

Example 5: Q_8

- **Generators:** $S = \{i, -i, j, -j\}$
- **Relator:** $ij(-i)j = 1$
- **Stats:** $V = 8, E = 16, F = 8$
- **Result:** $\chi = 0$ (**Torus**)

Comparison Table

Group	Generators	Based cycle / relator	V	E	F	χ
D_4 (cube)	$\{r, r^{-1}, s\}$	$r^4, (rs)^2$	8	12	6	2
D_4 (hex)	$\{r, r^{-1}, s\}$	$(1, r, r^2, r^2s, rs, s, 1)$	8	12	4	0
S_3	$\{s, t\}$	$(st)^3$	6	6	2	2
S_4	$\{s_1, s_2, s_3\}$	$(s_1s_3)^2, (s_1s_2)^3, (s_2s_3)^3$	24	36	14	2
Q_8	$\{i, -i, j, -j\}$	$ij(-i)j$	8	16	8	0